

**RSA****What is RSA?**

RSA is an encryption algorithm. It is widely used for encrypting important messages or digitally signing documents for e-commerce, and is included as part of the Web browsers by Netscape and Microsoft.

Who developed RSA?

The encryption algorithm was developed by Ron Rivest, Adi Shamir and Len Adleman (R, S and A) at the Massachusetts Institute of Technology (MIT) in 1977.

How does it work?

RSA encrypts on the principle of a private key and a public key. Users who wish to encrypt data first encrypt the message using a private key. This encrypted message can only be decrypted using a public key, which has already been distributed to the recipients of the message. The primary advantage of RSA comes from the fact that, while it is relatively easy to multiply two huge prime numbers together to obtain their product, it is computationally difficult to do the reverse.

Where does RSA stand today?

RSA is still the most widely used encryption algorithm. However, while other standards such as DES are faster to decrypt, RSA remains an industry favourite for encrypting data, with many believing that its 2048-bit key encryption is virtually unbreakable.

Why use an encryption algorithm at all?

With so much electronic data interchange (EDI) taking place in the form of e-commerce, it's essential to secure critical data. Several privacy laws make it mandatory that personal information of customers and its integrity is not compromised.

When will RSA be cracked?

Although RSA securities claim that the 2048-bit key encryption cannot be cracked, they offer bounties of up to \$200,000 (Rs 87 lakh) to anyone who can crack their algorithm.

THE VISUAL PRISM

SGI Introduces Linux-based Visual System

Silicon Graphics Systems India (SGI) launched the Linux-based visual computer product line, the Silicon Graphics Prism family of visualisation systems in India. With the Silicon Graphics Prism, SGI becomes the only hardware company in the world to offer a visual computer product line, based completely on open standards with Linux OS and commercial off-the-shelf components such as Intel Itanium 2 processors and graphics accelerators from ATI.

Scaling from one to 16 graphics pipelines and from one to 256 processors, the Silicon Graphics Prism family offers many times the visualisation capacity of any currently available computing system. For users, this scalability translates into the ability to interactively visualise terabytes of data in their native form without spending time culling it beforehand.

PROTECT YOUR SYSTEM

HP Launches Desktop With Security Features

Designed to help protect customers' business systems, the new HP Compaq dc7600 Business Desktop will be the first to ship standard with the HP ProtectTools embedded security chip. Developed by

Broadcom, the chip provides customers with hardware-based encryption, enhanced data protection and system access.

The new chip allows HP to deliver the first business desktop compatible with the future security features expected in Microsoft's Longhorn OS.

The dc7600's customised thermal design also helps enhance productivity and minimises system downtime by providing customers more reliable systems. The airflow design is structured to help cool all components of the desktop, as increasing demands for power are required.

In addition, the dc7600 series is the first desktop PC from HP to fully meet the European Union's Restrictions on Hazardous Substances directive. All PCs sold in EMEA must meet these requirements by July 2006 and HP is moving quickly to make it easy for customers to plan their desktop deployments.

NEW CHIP ON THE BLOCK

AMD Launches Multi-Core Opteron Processor

AMD has launched its dual-core 64-bit Opteron processor for servers and workstation PCs in India almost a month after its launch in America. According to AMD the new processors will improve performance by almost 90 percent as compared to single-core Opteron processor.

With the Direct Connect Architecture AMD has eliminated the Front Side Bus Architecture, thus eliminating the data transfer bottlenecks. The processors were designed from ground up to directly

connect two cores on a single die, along with memory, I/O and dedicated caches. Each of these CPU core has its own 1 MB L2 cache. An integrated DDR DRAM memory controller decreases the memory latency and increases performance power. With support to up to three coherent HyperTransport Technology links each processor provides a peak I/O bandwidth of up to 24 Gbps. The company also claims that the processor provides the best performance-per-watt ratio. The non-disruptive upgrade path of the Opteron is



another elegant feature that will attract consumers, says the company. The dual-core processor will work just as fine with the same infrastructure as a single-core AMD 64 processor.

Speaking at the launch, Ajay Marathe, President AMD India said that the simple out of the box offering ensured that the users would not have to undertake huge upgradation costs while moving from a single-core to a dual-core processor. The new processor also supports the present 64-bit and 32-bit software available in the market.

Offered in three series, 100, 200 and 800, the Dual-Core AMD Opteron 800 and 200 series has been available since May.

AMD also announced that the AMD Athlon 64 X2 processor will be launched in June this year. ■

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